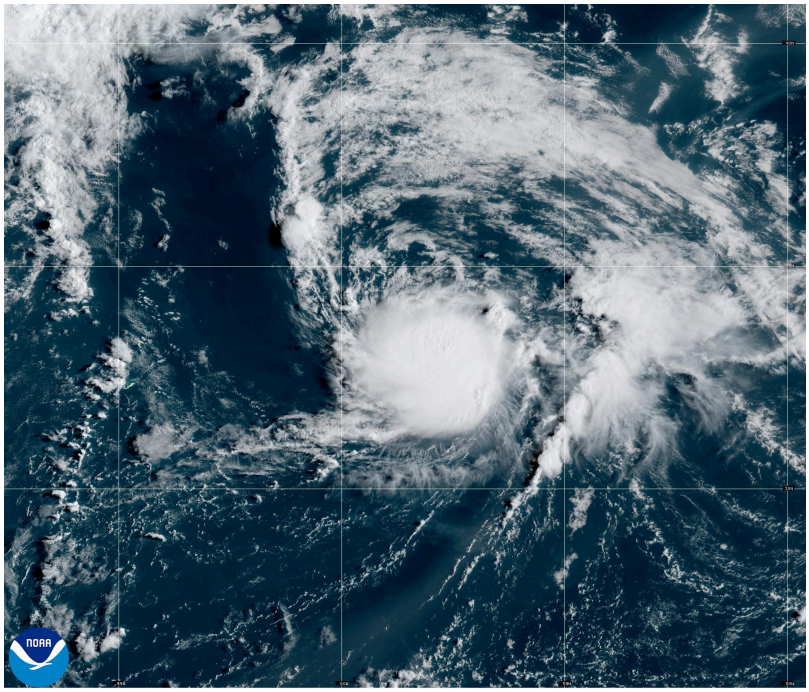
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

TROPICAL STORM FERNAND

(AL062025)

23–27 August 2025

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National Hurricane Center
13 November 2025



GOES-19 GEOCOLOR SATELLITE IMAGE OF TROPICAL STORM FERNAND AT 1200 UTC 25 AUGUST. IMAGE COURTESY OF NOAA/NESDIS/STAR.

Fernand was a tropical storm over the western Atlantic that did not affect land.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Fernand.

Tropical Storm Fernand

23–27 August 2025

BEST TRACK

The "best track[2]" positions and intensities for Tropical Storm Fernand are listed in Table 1. The "best track" chart of Fernand's path is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively. Observations also include flight-level measurements from 1 flight of the 53rd Weather Reconnaissance Squadron of the U.S. Air Force Reserve Command.

There were no ship reports of winds of tropical storm force associated with Fernand.

Origin

Fernand developed from a portion of a tropical wave that moved off the west coast of Africa on 14–15 August. The wave moved westward across the tropical Atlantic Ocean for 7 days. A few hundred miles east of the Lesser Antilles on 21–22 August, the northern portion of the wave turned northwestward to northward and became more convectively organized. Around 1800 UTC 23 August, Air Force Hurricane Hunter aircraft data indicated that the system had a well-defined circulation marking the formation of a tropical storm located about 465 n mi south-southeast of Bermuda.

Peak Intensity and Minimum Pressure

Fernand's peak intensity of 50 kt is supported by an ASCAT pass from 1343 UTC 25 August that had a peak vector of 48 kt. The minimum central pressure of 999 mb is primarily based on the Knaff-Zehr-Courtney pressure-wind relationship.

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[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Fernand.

CASUALTY AND DAMAGE STATISTICS

There were no reports of damage or casualties associated with Fernand.

FORECAST AND WARNING VERIFICATION

Table 2 provides the number of hours in advance of formation with the first NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. The formation of Fernand was generally well forecast, with the system being introduced into the 7-day TWO 162 hours before genesis. Probabilities for the 2-day TWO were introduced 120 hours before development. Figure 4 shows composites of 7-day TWO genesis areas for each category prior to the formation of Fernand. The initial genesis areas did not encompass the ultimate formation location. Instead they highlighted the area east of the northern Leeward Islands, suggesting early model guidance and official forecasts expected Fernand to form earlier than it did. Once the probabilities were raised to the high category (>60%) in the 7-day TWO 66 hours before development, the genesis areas consistently encompassed the formation location.

A verification of NHC official track forecasts for Fernand is given in Table 3a. Official track forecast errors were significantly greater than the mean official errors for the previous 5-yr period. The NHC errors were also greater than the climatology/persistence model (OCD5) for the 36–72 hour forecast periods, indicating that the official prediction lacked skill during that time period. The main factor contributing to the high errors was a noticeable northward bias of the official track prediction (not shown). A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. Many of the various individual models and consensus aids had lower track errors, especially beyond the 24-h mark. The Google DeepMind ensemble mean (GDMI) had the lowest error for most of the verifying forecast hours and was comparable to the best performing aids at 60 and 72 hours. It should be noted that GDMI was an experimental (non-operational) aid and not always available to the forecasters in real-time.

A verification of NHC official intensity forecasts for Fernand is given in Table 4a. Official intensity forecast errors were lower than the mean official errors for the previous 5-yr period. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b. The official forecast generally had some of the lowest errors compared to the intensity models throughout the forecast period, except for 48 h.

There were no coastal watches or warnings issued for Fernand.

Table 1. Best track for Tropical Storm Fernand, 23–27 August 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
23 / 1800	26.6	61.7	1010	35	tropical storm
24 / 0000	28.0	61.3	1009	35	"
24 / 0600	29.2	60.9	1009	35	"
24 / 1200	30.3	60.4	1009	35	"
24 / 1800	31.3	60.0	1006	40	"
25 / 0000	32.1	59.6	1005	45	"
25 / 0600	33.0	59.0	1004	45	"
25 / 1200	33.8	58.4	999	50	"
25 / 1800	34.8	57.6	999	50	"
26 / 0000	35.6	56.8	1002	45	"
26 / 0600	36.4	55.9	1005	40	"
26 / 1200	37.3	55.1	1006	35	"
26 / 1800	37.8	54.0	1004	35	"
27 / 0000	38.2	52.7	1002	40	"
27 / 0600	38.3	51.4	1000	45	"
27 / 1200	38.2	50.3	1000	45	"
27 / 1800	38.9	48.4	1000	45	low
28 / 0000	39.7	46.3	1002	45	"
28 / 0600	40.6	44.2	1006	40	"
28 / 1200	41.8	42.0	1007	35	"
28 / 1800	43.0	39.1	1008	35	"
29 / 0000					dissipation
25 / 1200	33.8	58.4	999	50	maximum winds and minimum pressure

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	120	162
Medium (40%-60%)	54	138
High (>60%)	36	66

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Tropical Storm Fernand, 23–27 August 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	27.1	62.1	113.1	183.6	234.3	323.6		
OCD5	34.7	65.7	84.2	95.8	111.6	123.3		
Forecasts	14	12	10	8	6	4		
OFCL (2020-24)	23.0	34.3	45.8	58.7	73.5	89.8	128.7	185.4
OCD5 (2020-24)	45.1	95.7	150.9	203.1	252.7	295.4	366.2	426.6

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Tropical Storm Fernand, 23–27 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	20.7	51.1	108.1	140.2	196.6	329.8		
OCD5	24.5	**49.3**	**73.5**	**87.7**	**98.4**	**100.7**		
GFSI	33.0	76.3	155.8	165.4	**176.3**	**295.3**		
HWFI	29.1	60.7	130.1	176.2	237.4	368.6		
HMNI	30.7	62.7	136.7	195.5	244.8	352.6		
HFAI	29.5	55.5	123.8	**136.7**	**174.4**	**292.9**		
HFBI	30.2	64.9	137.8	**125.1**	**138.1**	**198.5**		
EMXI	22.4	**45.1**	**91.3**	**106.2**	**158.4**	**288.2**		
CMCI	36.9	79.5	155.0	215.7	272.8	435.4		
CTCI	32.5	70.4	131.1	**130.0**	**138.7**	**187.8**		
TVCN	26.2	60.6	121.8	141.7	**180.6**	**283.5**		
TVCX	27.7	59.4	123.7	142.6	**181.8**	**283.5**		
GFEX	27.0	58.9	121.8	**136.9**	**167.5**	**291.5**		
TVDG	28.3	60.5	128.0	148.2	**183.9**	**283.5**		
HCCA	27.0	63.4	129.1	153.0	208.7	342.6		
FSSE	29.4	62.3	127.2	145.0	**196.3**	**322.1**		
AEMI	32.6	75.4	143.7	145.9	**159.9**	**194.3**		
GDMI	**11.2**	**21.5**	**30.2**	**32.8**	**67.2**	**104.2**		
TABS	23.9	53.8	**98.2**	**116.4**	**153.2**	**257.0**		
TABM	**18.1**	**40.8**	**67.8**	**56.8**	**60.2**	**127.7**		
TABD	**15.1**	**30.2**	**52.9**	**63.7**	**96.8**	**89.6**		
Forecasts	9	8	8	5	3	2		

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Tropical Storm Fernand, 23–27 August 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**3.6**	**4.2**	**5.5**	**8.8**	**10.0**	**8.8**		
OCD5	5.1	9.7	10.9	13.4	15.2	13.0		
Forecasts	14	12	10	8	6	4		
OFCL (2020-24)	5.1	7.3	8.6	10.0	10.5	10.9	12.4	13.6
OCD5 (2020-24)	6.8	10.6	13.8	16.5	17.9	19.2	21.4	19.9

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Tropical Storm Fernand, 23–27 August 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	2.8	2.8	5.0	8.0	8.8	6.7		
OCD5	5.1	9.4	11.4	14.6	14.2	10.0		
HWFI	4.1	6.1	7.2	**1.6**	**6.2**	7.7		
HMNI	4.9	4.6	5.4	**4.2**	10.0	15.0		
HFAI	4.0	3.7	6.3	**3.8**	9.5	8.7		
HFBI	5.2	4.7	**3.7**	**6.0**	9.8	9.7		
DSHP	4.0	4.9	**4.3**	**3.4**	**7.8**	**5.7**		
LGEM	4.4	6.6	6.7	**2.6**	**5.5**	10.3		
ICON	3.8	4.4	5.4	**1.6**	**7.2**	8.7		
IVCN	3.9	3.3	5.0	**3.2**	**8.2**	8.7		
IVDR	3.8	3.1	5.1	**3.0**	**8.0**	9.0		
CTCI	3.4	5.1	8.3	**6.8**	10.0	10.7		
GFSI	3.8	5.3	7.1	**2.2**	**4.8**	10.0		
EMXI	5.3	8.0	7.4	**3.4**	**5.5**	**5.7**		
HCCA	3.7	3.3	5.4	**4.6**	8.8	7.7		
FSSE	3.7	3.4	5.7	**4.4**	9.0	9.3		
GDMI	3.8	3.3	**4.4**	**6.8**	**6.8**	**5.0**		
Forecasts	9	9	9	5	4	3		

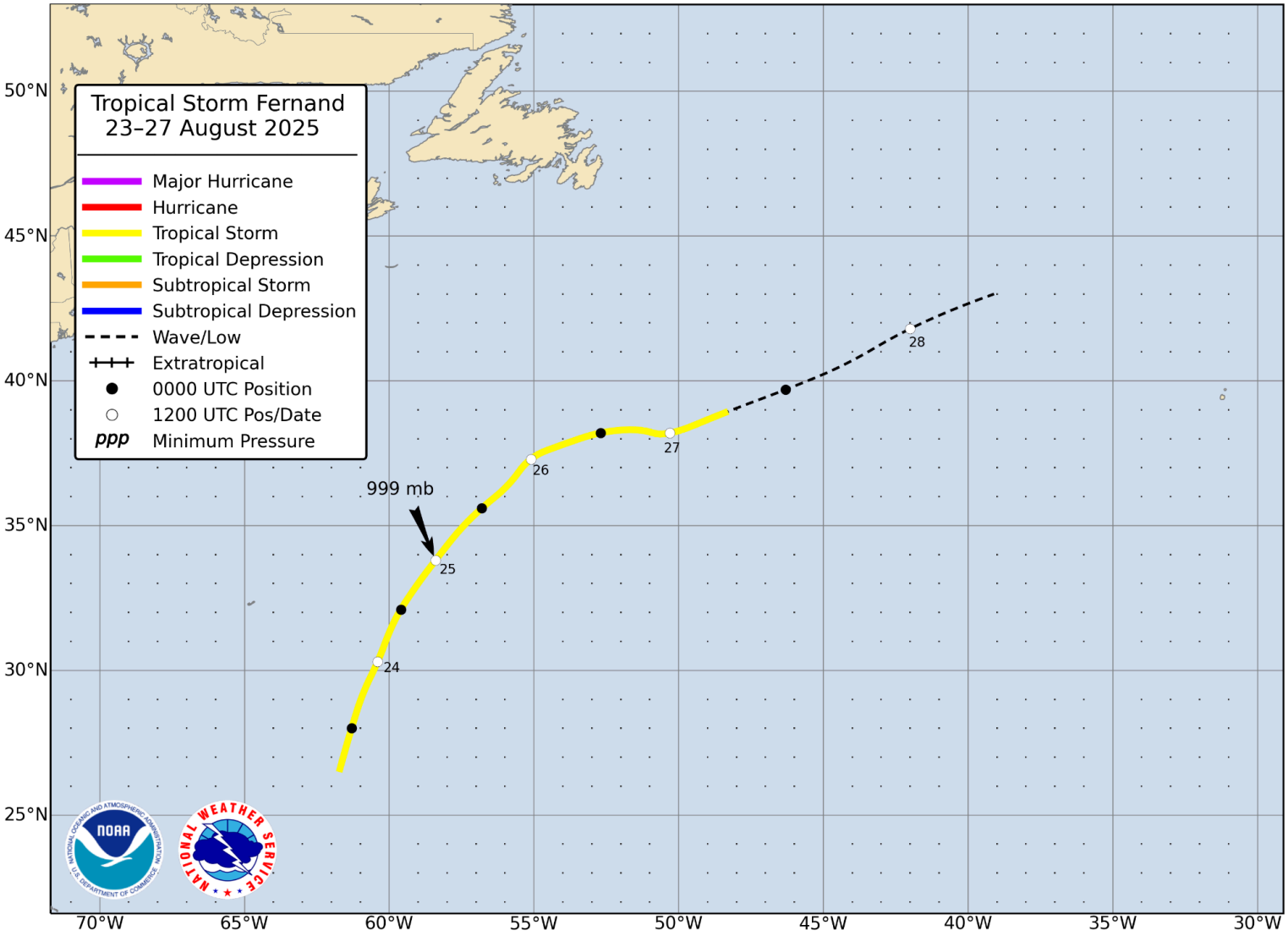


Figure 1. Best track positions for Tropical Storm Fernand, 23–27 August 2025.

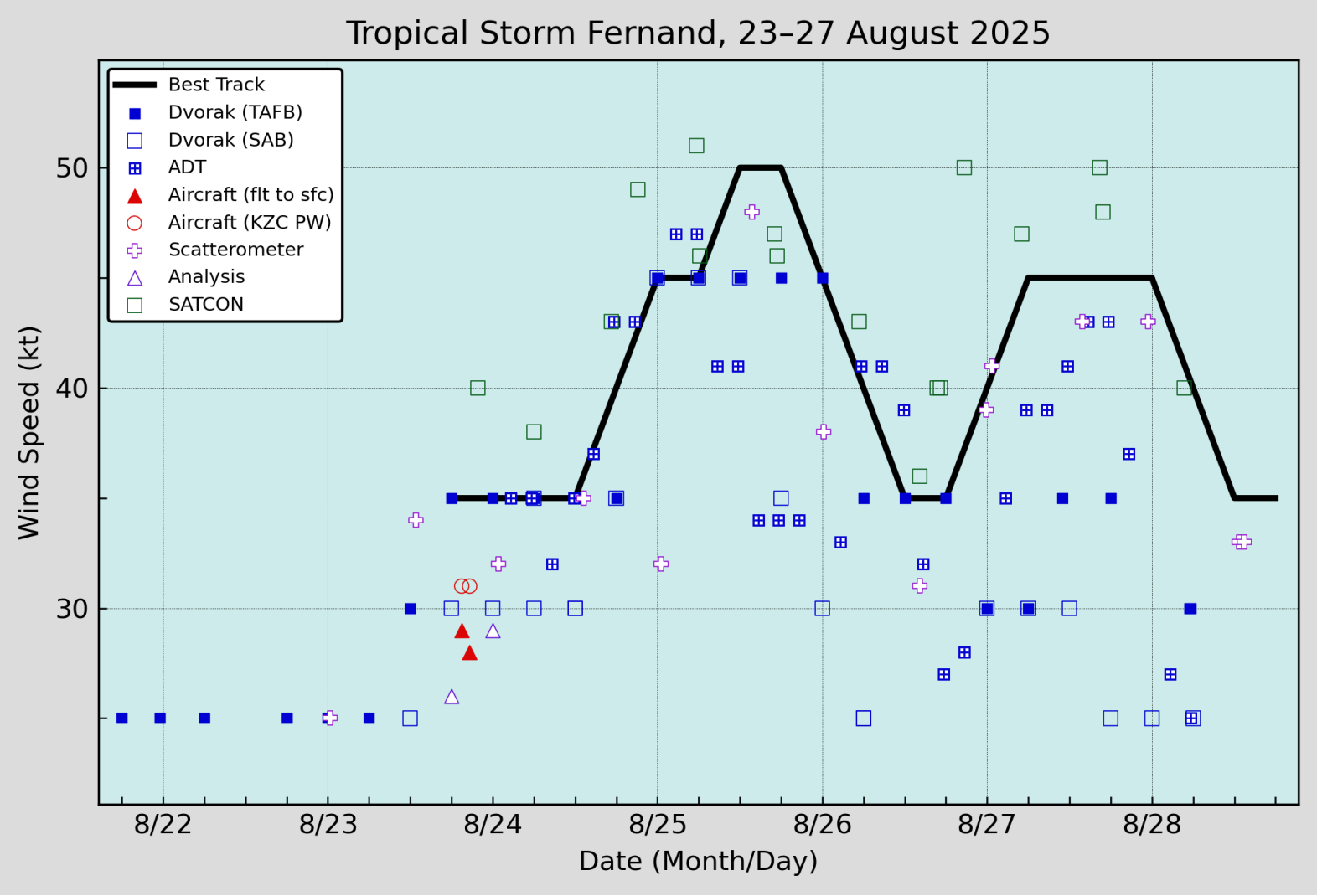


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Tropical Storm Fernand, 23–27 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

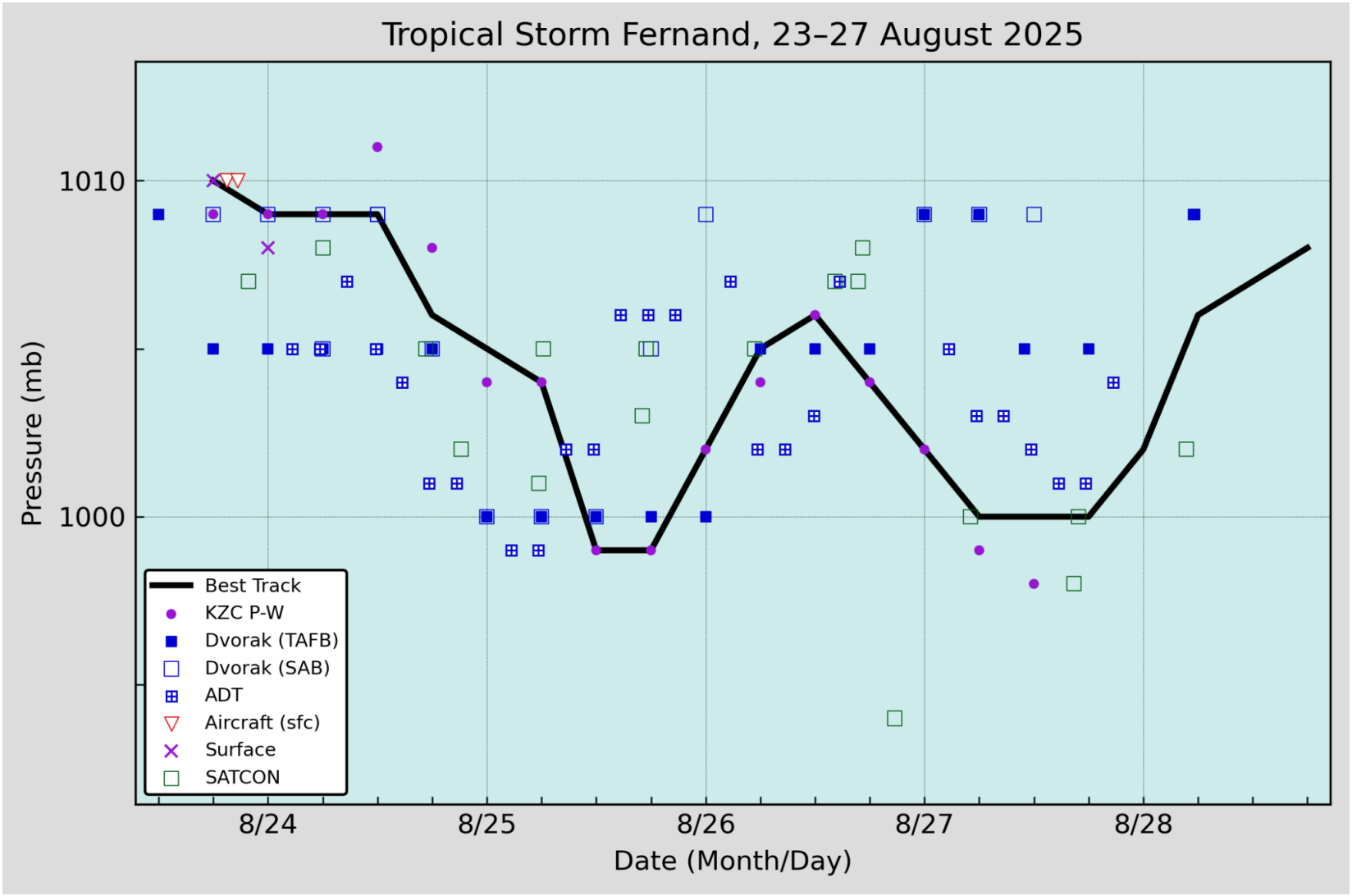


Figure 3. Selected pressure observations and best track minimum central pressure curve for Tropical Storm Fernand, 23–27 August 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

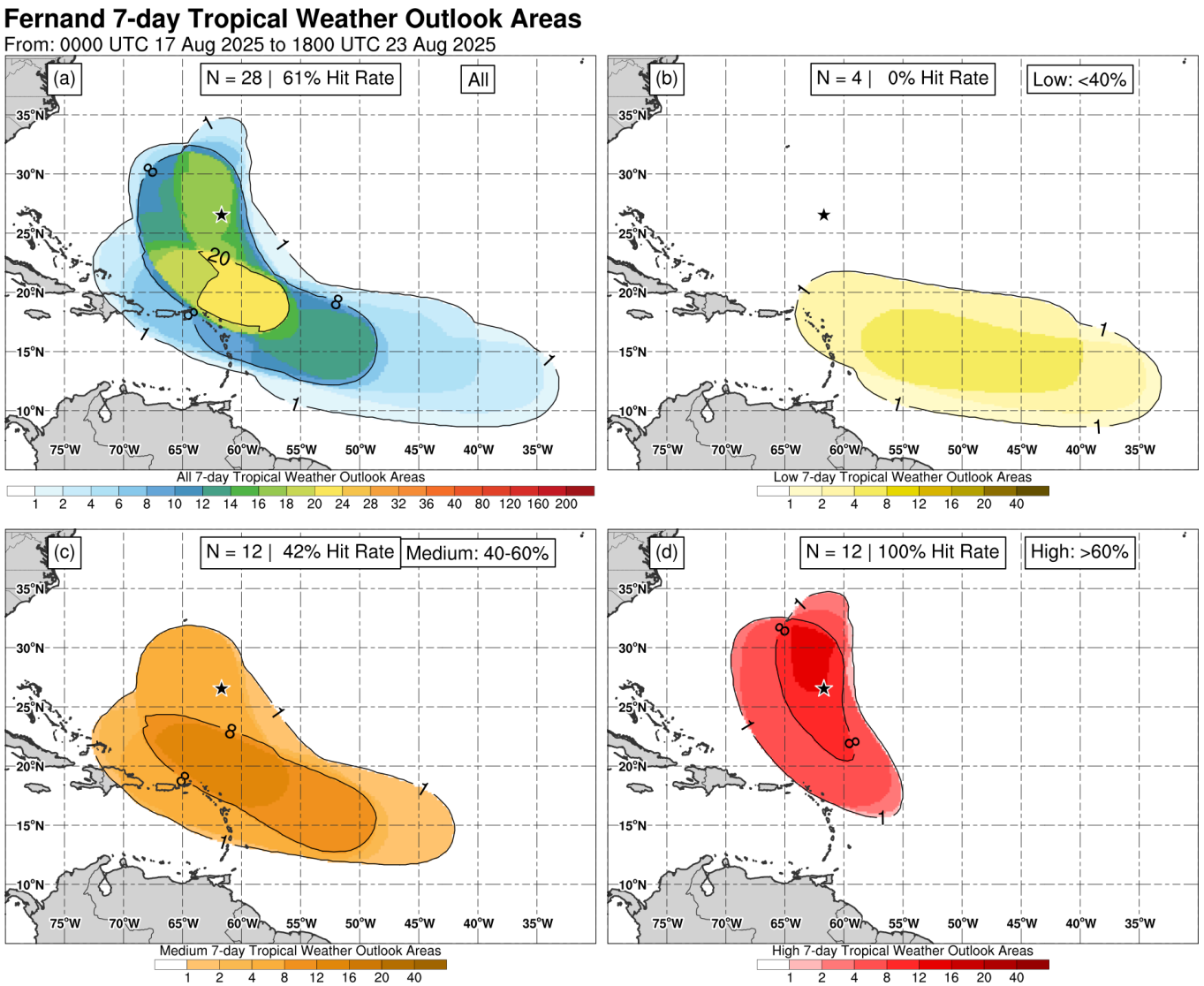


Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Fernand for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.